

GENERAL

ALIAPT APT3100 is a high performance compact pressure transmitter with a wide range of available measuring ranges, process connections, output signals, and electrical connections. It features high stability, strong reliability and anti-interference.

The model APT3100 for general industrial applications is the ideal solution for customers with demanding measuring requirements.

FEATURES

- ❑ Dedicated high-performance ASIC regulating circuit
- ❑ Short response time
- ❑ Measuring ranges from 0-0.1~0-600 bar
- ❑ Output signal: 4-20 mA (2-Wire) or 1-5 VDC
- ❑ High accuracy of +/-0.20% of span
- ❑ Compact size to meet the needs of small space applications
- ❑ Multiple configurations to choose from for the best solution

STANDARD SPECIFICATION

- Process Fluid : Liquid, Gas, Vapor
- Application : Absolute Pressure, Gauge Pressure, Vacuum
- Measuring Range : 0-0.1~0-600 bar
- Accuracy : +/-0.45% of span (Standard)
: +/-0.20% of span (Option)
- Non-Linearity : ≤0.2% of span (BFSL) (Standard)
: ≤0.1% of span (BFSL) (Option)
- Non-Repeatability : ≤0.05% of span
- Stability : ≤+/-0.2% of URL for 1 year
- Working Temperature : -40~85 °C
- Compensation Temp. : -10~70 °C
- Temp. Effect of zero
Ranges≤0.25 bar : ≤0.4% FS / 10K
Ranges>0.25 bar : ≤0.2% FS / 10K
- Temp. Effect of span : ≤0.2% FS / 10K
- Over Pressure Limit : 250% of Full Scale
- Vibration Resistance : 20 g (20-2000 Hz) per IEC 60068-2-6
- Shock Resistance : 100 g (10 ms)
- EMI / RFI Effect : Follow SAMA PMC 33.1 from 20-1000 MHz and for field strengths up to 30 V/m
- Ambient Temperature : -20~70 °C
- Humidity Limit : 0-95% Relative Humidity
- Process Connection : G1/4" (Standard)
M12 * 1.5, 1/4"-18 NPTM, 1/8"-27 NPTM,
7/16"-20 UNF (Female) etc. (Option)

- Pressure Port : 3.0 mm
- Response Time : ≥1 ms
- Output Response Time : ≤8 ms
- Current Output : 4-20 mA (2-Wire)
Load : ≤ (VDC-9) / 0.02A
- Voltage Output : 1-5 VDC (3-Wire)
Load : > 5000 ohm
- Power Supply : 9-32 VDC
- Material
Connection : Stainless steel 304
Diaphragm : Stainless steel 316L
Body : Stainless steel 304
Fill Fluid : Silicone Oil
Wetted O-ring (inside) : Viton
Seal Gasket (outside) : NBR
- Electrical Connector
Cable Outlet : 2/3-Cable, Standard Outlet Cable 0.5 m
Packard : 3-Pin, Standard Outlet Cable 1 m
DIN EN175301-803A : 3-Pin, PG9, Without Outlet Cable
M12 * 1 : 4-Pin, Standard Outlet Cable 1.5 m
- Protection Class : IP54 (Packard)
: IP65
- Intrinsically Safe : Ex ia IIC T6 (Standard)
- Weight : Approx. 80 g



DIMENSIONS

DIN EN175301-803A		M12 * 1		Cable Outlet		Packard	
4-20mA(2-Wire)	1-5VDC(3-Wire)	4-20mA(2-Wire)	1-5VDC(3-Wire)	4-20mA(2-Wire)	1-5VDC(3-Wire)	4-20mA(2-Wire)	1-5VDC(3-Wire)
Pin1: VCC Pin2: GND Pin3: N/A	Pin1: VCC Pin2: GND Pin3: Output	Pin1: VCC Pin2: GND Pin3: Output	Pin1: VCC Pin2: GND Pin3: Output	Red: VCC Black: GND	Red: VCC Black: GND Yellow: Output	Pin1: VCC Pin2: GND Pin3: N/A	Pin1: VCC Pin2: GND Pin3: Output

MEASURING RANGE

Range Code	Pressure Range	Range Code	Pressure Range	Range Code	Pressure Range	Range Code	Pressure Range
0	0-0.1 bar	5	0-1 bar	A	0-10 bar	F	0-100 bar
1	0-0.16 bar	6	0-1.6 bar	B	0-16 bar	G	0-160 bar
2	0-0.25 bar	7	0-2.5 bar	C	0-25 bar	H	0-250 bar
3	0-0.4 bar	8	0-4 bar	D	0-40 bar	J	0-400 bar
4	0-0.6 bar	9	0-6 bar	E	0-60 bar	K	0-600 bar

MODEL SELECTION GUIDE

APT3100 Series							
Example: APT3100-G9NN-NNNNN-NNN							
APT3100-	X	X	X	X	-XXXXX	-XXX	Description
Type	G						Gauge Pressure Transmitter
	A						Absolute Pressure Transmitter * Note 1
Pressure Range	0-K						Please see MEASURING RANGE
Electrical Connector	N						L-connector DIN EN175301-803A (Without Outlet Cable)
	M						180° Round connector M12 * 1 (Standard Outlet Cable 1.5 m)
	R						90° Round connector M12 * 1 (Standard Outlet Cable 1.5 m)
	C						Cable outlet (Standard Outlet Cable 0.5 m)
	P						Packard IP54 (Standard Outlet Cable 1 m)
Output	N						4-20 mA (2-Wire)
	V						1-5 VDC (3-Wire)
Process Connection	-NNNNN						G1/4" (Standard)
	-XXXXX						1/4"-18 NPTM, M12 * 1.5, 7/16"-20 UNF etc. (With Adapter)
Option	-NNN						None
	-020						Accuracy +/-0.20% of span * Note 2

Note 1: Absolute Pressure Transmitter only for range code 5-B

Note 2: Accuracy +/-0.20% of span only for range code 3-K